

*Neural Mechanisms of Reinforcement Learning: Developmental Variance in Reward
Prediction Error Signalling and Adolescent Risk-Taking.*

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ABSTRACT

Adolescence is associated with elevated risk-taking behaviour, which has been linked to immature reward processing in subcortical brain regions, particularly the ventral striatum. Although reward prediction error signalling and adolescent risk-taking have been examined separately, relatively few studies have directly integrated developmental differences in ventral striatal RPE signals with behavioural risk-taking outcomes. The present study examines the extent to which age-related differences in ventral striatal reward prediction error signals explain elevated risk-taking behaviour in adolescents compared to adults. A literature review was combined with a partial analysis of published summary statistics from Vaidya et al. (2013), comparing fMRI responses to monetary rewards across 18 adolescents and 18 young adults. Demographic analysis of Smith et al. (2024) dataset was conducted to contextualise lifespan coverage across 114 participants aged 21-80. Adolescents showed attenuated ventral striatal sensitivity to reward magnitude differences relative to adults, reflected in a significant reward magnitude by age group interaction in right VS: $F(2, 68) = 4.27, p = 0.02$. Adults demonstrated contextual recalibration of VS reward signals depending on relative reward value, a pattern absent in adolescents. Lower VS differentiation was independently associated with higher

impulsivity scores across both groups, and adolescents scored significantly higher on impulsivity than adults. These findings partially support the dual-systems model, suggesting adolescent risk-taking reflects an immature RPE signalling system characterised by reduced magnitude sensitivity and context-insensitive reward encoding. Causal inference is limited by the cross-sectional design and single primary source, and future longitudinal research with larger and more diverse samples is needed.

INTRODUCTION:

To better understand this topic, it is necessary to understand how the ventral striatum supports reward processing, what reward prediction errors (RPEs) are within reinforcement learning models, and how adolescent brain development differs from that of adults—particularly in regions governing reward sensitivity and cognitive control. Reward prediction errors refer to the difference between expected and received outcomes. For example, receiving an unexpectedly large reward produces a positive prediction error, while failing to receive an anticipated reward produces a negative prediction error. Adolescence represents a period of heightened neural plasticity and behavioural experimentation, making it especially relevant for studying reinforcement learning mechanisms. This topic is of importance in neuroscience because altered RPE signalling during adolescence contributes to heightened reward sensitivity and risk-taking behaviour, with direct consequences including substance misuse, unsafe driving, financial instability, and other adverse outcomes. Although individual studies examine adolescent risk-taking and others model RPE signals in the ventral striatum (Cohen et al., 2010; Van Leijenhorst et al., 2009), relatively few studies have integrated these developmental neural mechanisms with risk-taking behavioural outcomes across age groups. This leaves it unclear whether developmental differences in RPE signalling underlie the elevated risk-taking observed

in adolescents. The argument made in the present study is that developmental differences in ventral striatal reward prediction error signalling partially explain elevated adolescent risk-taking through reduced reward magnitude sensitivity and weaker contextual reward recalibration. These findings may also carry broader implications for quantitative models that incorporate RPE-driven decision-making. The question the present study seeks to answer is **to what extent age-related differences in ventral striatal reward prediction error signals explain elevated risk-taking behaviour in adolescents compared to adults?**

REWARD PREDICTION ERROR SIGNALLING IN THE VENTRAL STRIATUM.

The ventral striatum and the nucleus accumbens play a key role in encoding reward prediction errors, which are the differences between expected and received outcomes that underlie reinforcement learning. Foundational work by Schultz et al. (1997) established that midbrain dopamine neurons fire in response to unexpected rewards and suppress activity when expected rewards are omitted, providing the biological foundation for RPE computation. Subsequent human neuroimaging work confirmed these findings, with blood-oxygen-level-dependent (BOLD) responses in the ventral striatum tracking RPEs during reward learning tasks. Positive prediction errors produced robust activation, and negative errors produced suppression (Cohen et al., 2010). Garrison et al. (2013) used a meta-analysis to show that striatal activity encodes prediction errors across reinforcement learning paradigms, highlighting its important role in updating value representations.

ADOLESCENT BRAIN DEVELOPMENT AND REWARD SENSITIVITY.

Adolescence is characterised by a well-documented developmental imbalance between subcortical reward circuitry, which matures relatively early, and prefrontal regulatory regions,

which continue developing into the mid-twenties. Steinberg (2008) described this as a dual-systems model, arguing that elevated reward sensitivity driven by limbic system maturation outpaces the development of cognitive control. This developmental imbalance produces a window of heightened impulsivity and risk-taking, which can often explain why adolescents tend to make more suboptimal decisions under uncertainty when compared to adults. Structural and functional neuroimaging supports this account: Casey et al. (2008) reviewed and synthesised evidence suggesting that adolescents show greater amygdala and ventral striatum reactivity to emotional and rewarding stimuli compared to both children and adults, reflecting a non-linear developmental trajectory. Van Leijenhorst et al. (2009) found that ventral striatal responses to reward outcomes peak in adolescence with more complex patterns for anticipation, which suggests non-linear developmental trajectories in reward processing. Although influential, Steinberg's dual-systems model has also been criticised for oversimplifying developmental variability and underestimating individual differences in adolescent neurodevelopment.

RPE SIGNALS AND RISK-TAKING BEHAVIOUR ACROSS DEVELOPMENT.

Neuroimaging studies consistently show that adolescents exhibit exaggerated ventral striatum activation relative to adults during reward anticipation, even when controlling for objective reward magnitude (Galván et al., 2006; Van Leijenhorst et al., 2009). This heightened neural response is consistent with an enhanced reward sensitivity and altered RPE signal encoding that biases adolescents toward options with uncertain but potentially large payoffs, which is a pattern Paulsen et al. (2012) link directly to elevated risk-taking on behavioural tasks. Critically, this striatal hyperreactivity co-occurs with protracted maturation of prefrontal regulatory regions, meaning the cognitive control mechanisms that would ordinarily dampen RPE-driven impulses are not yet fully engaged (Casey et al., 2008; Steinberg, 2008). Together,

these findings suggest that adolescent risk-taking is grounded in a developmental mismatch between an RPE system that overweights positive outcomes and a regulatory system that has not yet caught up. However, there are still some inconsistencies in research, where some MID-task studies found reduced adolescent VS anticipation (like Bjork et al.) and others found hyperactivation depending on the circumstances.

IMPLICATIONS FOR REAL-WORLD DECISION-MAKING.

The neural and behavioural patterns described above have meaningful consequences beyond just the field of research. Spear (2000) identified adolescence as a sensitive period for substance initiation, arguing that heightened reward sensitivity lowers the threshold for drug reinforcement and escalation. Reyna and Farley (2006) complemented this by suggesting that adolescents rely more heavily on outcome-based reasoning. This is consistent with RPE-driven learning, as opposed to the gist-based heuristics that often protect adults from high-risk choices. Ernst et al. (2006) proposed a triadic model integrating approach, avoidance, and regulatory systems to explain why adolescents are simultaneously drawn to novelty and poorly equipped to inhibit risky behaviour. Collectively, this literature suggests that developmental RPE differences are not a narrow laboratory phenomenon but a mechanism with broad relevance to adolescent health and decision-making outcomes in the real world.

METHODOLOGY:

Given these theoretical inconsistencies and developmental findings, a structured secondary analysis was conducted to examine whether empirical evidence supports the proposed relationship between ventral striatal RPE signalling and adolescent risk-taking. This study utilises a structured literature review methodology combined with a descriptive secondary data synthesis to examine developmental differences in reward prediction error signalling and their

relationship to risk-taking behaviour across age groups. Sources were identified through database searches of APA PsycINFO, PubMed, Google Scholar, and released university papers. The search terms used included: “reward prediction error,” “ventral striatum,” “adolescent development,” and “risk-taking.” Inclusion criteria required that the studies be peer-reviewed, published in English, involve human participants, and directly measure either neural RPE signals or behavioural risk-taking outcomes across different age groups. Studies relying on animals, lacking peer review, or using samples below 10 participants were excluded. Included studies were analysed and grouped according to whether they address neural mechanisms, behavioural outcomes, or the relationship between the two. This approach allows for a structured synthesis of findings rather than a surface-level summary that reiterates what is already known.

In addition to the literature review, a descriptive secondary data analysis was conducted using published summary statistics from 2 complementary empirical studies. This involves extracting reported effect sizes, means, t-values, F-values, and correlation coefficients directly from published figures and tables, and then using these to conduct secondary, descriptive quantitative comparisons without requiring access to raw participant data. The primary source was Vaidya et al. (2013), published in PLOS ONE, and is titled “Neural Sensitivity to Absolute and Relative Anticipated Reward in Adolescents”. Vaidya et al. (2013) recruited 18 adolescents (mean age 13.39 years, SD = 0.92) and 18 young adults (mean age 27.72 years, SD = 1.36) completing a modified Monetary Incentive Delay (MID) task across 3 reward magnitudes (\$0.20, \$1.00, \$5.00) during fMRI scanning. The supplementary source was Smith et al. (2024), published in Data in Brief via NIH PMC PubMed Central, titled “Social reward and nonsocial reward processing across the adult lifespan: An interim multi-echo fMRI and diffusion dataset”. The supplementary source provides data from 114 participants aged 21-80 completing a monetary

reward anticipation paradigm, extending the developmental comparison through older adulthood. The Smith et al. (2024) dataset was included to contextualise the lifespan coverage of these findings through demographic analysis of its 114 participants aged 21-80. Neural analysis of the Smith dataset was not conducted, as the raw fMRI data requires specialist neuroimaging preprocessing software beyond the scope of this paper, and the social reward paradigms employed are not directly comparable to the monetary incentive delay task used by Vaidya et al. (2013). Demographic analysis of the Smith et al. (2024) participant data was conducted using Python (version 3.13.7), utilising the pandas and matplotlib libraries for data processing and visualisation.

Extracted variables include ventral striatal percentage signal change during reward anticipation as the primary neural-dependent measure, Spearman correlation coefficients between VS activation and Barratt Impulsiveness Scale scores as the behavioural-dependent measure, and peak t-values from whole-brain analyses as a secondary neural measure. Age group serves as the primary independent variable, operationalised as a three-level factor: adolescents (Vaidya, ages 12-15), young adults (Vaidya, ages 26-30; Smith, ages 21-35), and older adults (Smith, ages 50-80). Between-group comparisons of extracted means will assess whether adolescents show weakened VS magnitude sensitivity relative to adults, consistent with dual-systems model predictions. Extracted Spearman correlation coefficients will be compared across reward contrasts and impulsivity components to evaluate the relationship between VS differentiation and impulsive behaviour. When figures rather than tables were the primary data source, values were extracted from published graphs by visual inspection, with approximate estimates. All extracted values are reported alongside their original published statistics to ensure full reproducibility. Findings from both sources are compared against the developmental predictions outlined in the

accompanying literature review. When figures rather than tables were the primary data source, values were extracted by visual inspection with approximate estimates. These approximations were used only for descriptive comparison and not for novel inferential statistical testing.

RESULTS:

Sample Characteristics

The primary source, Vaidya et al. (2013), was composed of 18 adolescents (9 female, mean age 13.39 years, SD = 0.92, range 12-15) and 18 young adults (9 female, mean age 27.72 years, SD = 1.36, range 26-30), recruited from the University of Iowa community and matched on biological sex. The supplementary source, Smith et al. (2024), comprised 114 participants (70 female, 40 male, 4 other; mean age 42.78 years, SD = 17.13, range 21–80). Participants were categorised into three age groups: young adults aged 21–35 (n = 53, 46.5%), middle adults aged 36–49 (n = 21, 18.4%), and older adults aged 50–80 (n = 40, 35.1%). The sample was racially diverse, with 68 White participants (59.6%) and 46 non-White participants (40.4%), including Black or African American (n = 24, 21.1%), Asian (n = 11, 9.6%), and multiracial individuals (n = 8, 7.0%). Across both sources, the combined sample comprised 150 participants spanning ages 12 to 80, providing broad lifespan coverage for the developmental comparison. All Vaidya participants completed the modified MID task across 3 fMRI scanning runs. The following results are drawn from published summary statistics, figures, and tables reported in these two sources.

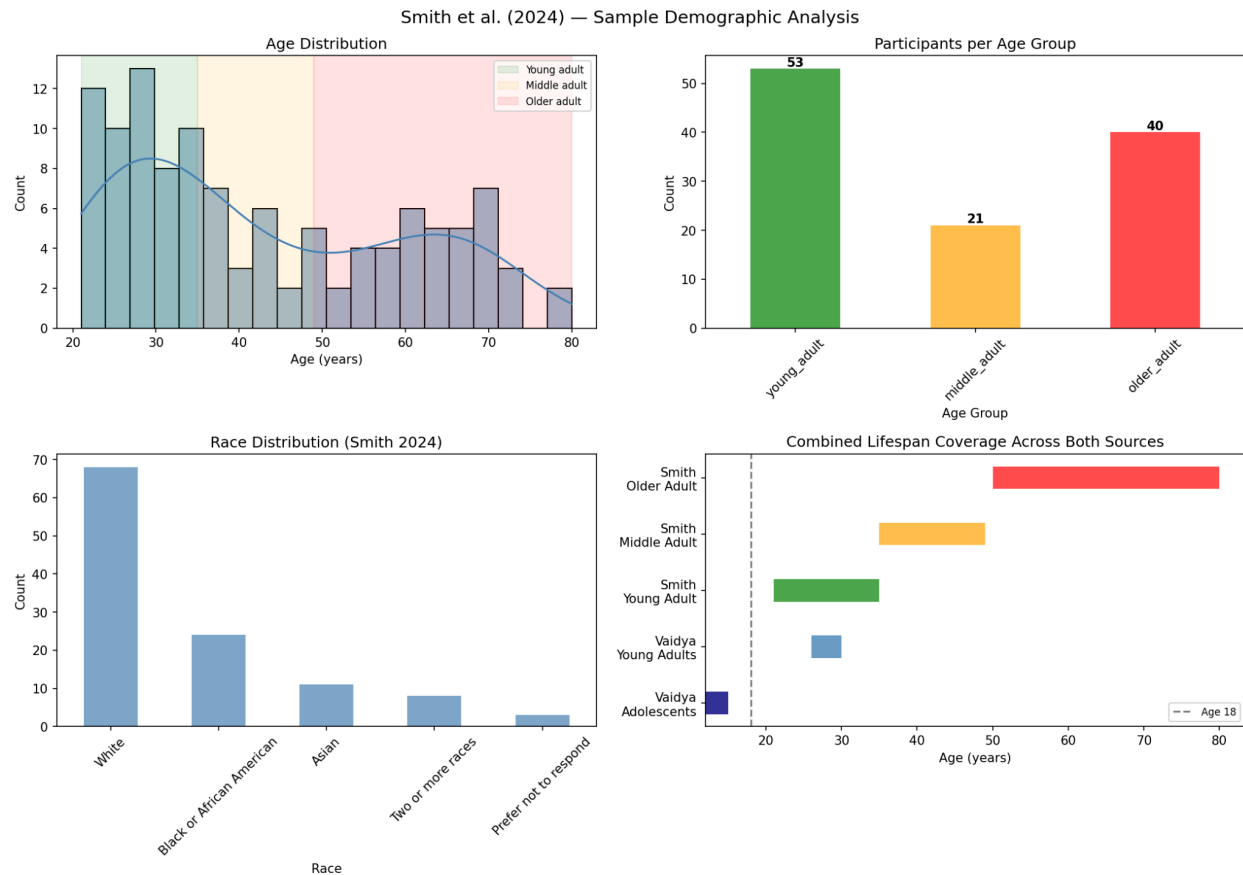


Figure 1

Behavioural Findings

Extracted behavioural data from Vaidya et al. (2013) indicate that adolescents and adults performed comparably on the MID task. No significant between-group difference was observed in hit rate ($p = 0.97$) or mean reaction time ($p = 0.80$), confirming that any subsequent neural differences cannot be attributed to differential task engagement or effort between groups. Both groups demonstrated faster reaction times and higher accuracy on higher reward trials, reflected in a significant main effect of reward magnitude on reaction time: $F(2, 68) = 19.12, p < 0.001$. This pattern of reward-based behavioural performance was consistent across age groups,

suggesting that both adolescents and adults were sensitive to reward magnitude at the behavioural level, even where neural sensitivity differed.

Primary Neural Finding: Absolute Reward Magnitude Processing

Extracted fMRI data from Vaidya et al. (2013) revealed a significant reward magnitude by age group interaction in the right ventral striatum: $F(2, 68) = 4.27, p = 0.02$. This result reflects a major difference in how the 2 groups encode increasing reward value. Adults demonstrated a mostly linear increase in VS BOLD activation across the \$0.20, \$1.00, and \$5.00 conditions, while adolescents showed a non-linear pattern in which VS activation did not scale proportionally with reward magnitude. Critically, there was no significant main effect of age group on overall VS activation ($p = 0.84$), indicating that the groups did not differ in their absolute level of VS response but rather in their sensitivity to reward magnitude differences. This finding suggests that developmental reward-processing differences may be qualitative rather than purely quantitative. The left VS showed the same pattern, but the result was not considered significant: $F(2, 68) = 2.71, p = 0.07$.

Whole-brain analyses further supported this dissociation. Adults showed significant activation in the VS, anterior cingulate cortex, thalamus, and insula during the absolute reward anticipation, as reported in Table 2 of Vaidya et al. (2013), with the largest cluster centred on the mesial occipital lobe, thalamus, and VS (peak $t = 9.39$, cluster size = 14,810 voxels). Adolescents showed no regions surviving the corrected activation threshold for the same contrast, suggesting a comparatively reduced whole-brain response to graded reward anticipation.

Secondary Neural Finding: Relative Reward Processing

Extracted data from Vaidya et al. (2013) further revealed a meaningful dissociation between age groups in relative reward processing. This was whether VS activation to the same

objective reward (\$1.00) differed depending on whether it represented the larger or smaller option within a given set. Among adults, VS activation to the \$1.00 reward was significantly greater when it was the larger of two options than when it was the smaller, in both the right VS: $t(17) = 3.09$, $p = 0.007$, and the left VS: $t(17) = 2.81$, $p = 0.01$. Adolescents showed no such context dependence: right VS: $t(17) = 0.98$, $p = 0.34$; left VS: $t(17) = 0.47$, $p = 0.65$. This pattern indicates that adults dynamically recalibrate their VS reward signal based on the relative value of options in context. This is consistent with sophisticated RPE computation, while adolescents encode reward in a more absolute, context-insensitive manner.

VS Impulsivity Correlations

Extracted correlation data from Table 1 of Vaidya et al. (2013) revealed that, after controlling for age group, lower VS differentiation between reward magnitudes was associated with higher impulsivity scores on both components of the Barratt Impulsiveness Scale. This distinction is especially important because it suggests that adolescent reward-processing differences reflect altered sensitivity and contextual calibration rather than simple hyperactivation. Specifically, the \$1.00 versus \$0.20 contrast in the right VS correlated negatively with attentional impulsivity (A_Imp): $r = -0.38$, $p < 0.05$, and S_Imp: $r = -0.34$, $p < 0.05$. These correlations were computed by the researchers using Spearman's rank-order method and controlled for age group, meaning they reflect a relationship between VS differentiation and impulsivity that holds across both adolescents and adults rather than being driven by group membership alone.

Adolescents scored significantly higher than adults on attentional impulsivity: $t(34) = -2.50$, $p = 0.02$, which is consistent with the dual-systems model prediction of elevated impulsivity during adolescence. Taken together, these findings suggest that reduced VS sensitivity to reward

magnitude differences (a pattern observed in adolescents) is independently associated with higher impulsivity, providing a neural mechanism through which developmental differences in RPE signalling may translate into elevated risk-taking behaviour.

DISCUSSION:

The findings extracted from Vaidya et al. (2013) provide partial but meaningful support in the answer to the research question of this study: age-related differences in ventral striatal reward prediction error signalling contribute to elevated risk-taking behaviour in adolescents relative to adults. Across three lines of evidence: absolute reward magnitude processing, relative reward context sensitivity, and the neural-behavioural correlation with impulsivity, there is a consistent pattern in which adolescents encode reward value in a qualitatively different manner than adults. This difference is independently associated with higher impulsivity scores. Impulsivity may represent one behavioural pathway through which altered RPE signalling contributes to elevated risk-taking. Although related, impulsivity and risk-taking are not identical constructs, and future studies should examine how these behavioural dimensions interact developmentally. The following discussion interprets these findings in relation to the dual-systems model, addresses inconsistencies in the broader literature, and considers the implications and limitations of the conclusions that can be drawn.

Interpretation of Primary Neural Findings

The most direct neural finding of this review is the significant reward magnitude by age group interaction in the right ventral striatum: $F(2, 68) = 4.27, p = 0.02$. Adults demonstrated an approximately linear increase in VS BOLD activation as reward magnitude increased from \$0.20 to \$1.00 to \$5.00, while adolescents showed no such proportional scaling. This is a theoretically important result because it suggests that adolescents do not encode the incremental value of

reward differences in the same way adults do. Within a reinforcement learning framework, this pattern is consistent with weakened RPE computation, where the VS is not updating its response in proportion to the magnitude of the prediction error signal. This is precisely what Schultz et al. (1997) identified as the core computational function of midbrain dopamine neurons. The absence of a significant main effect of age group ($p = 0.84$) strengthens this interpretation. It is not that adolescents show globally lower VS activation, but specifically that their activation does not differentiate between reward magnitudes. This distinction is critical because it means adolescents are not simply less motivated; adolescents are less sensitive to the relative size of potential gains, which has direct implications for how they evaluate risky options.

This finding aligns closely with Galván et al. (2006), who reported earlier maturation of the nucleus accumbens relative to the orbitofrontal cortex, and with Van Leijenhorst et al. (2009), who found non-linear developmental trajectories in VS reward processing. Both papers support the idea that the adolescent VS is not functioning in a manner equivalent to the adult VS, even where surface-level activation appears comparable. The present findings extend this by showing that the difference is specifically one of magnitude sensitivity rather than overall responsiveness.

Relative Reward Processing and RPE Sophistication

The relative reward findings provide a second and more nuanced line of evidence. Adults showed significantly greater VS activation to the \$1.00 reward when it was the largest available option compared to when it was the smallest: right VS $t(17) = 3.09$, $p = 0.007$; left VS $t(17) = 2.81$, $p = 0.01$. Adolescents did not show this environmental impact. This dissociation is consistent with the computational account of RPE signalling developed by Schultz (2000), in which prediction errors are calculated relative to available alternatives instead of against a fixed scale. Adult VS responses appear to incorporate this contextual recalibration, encoding reward

value as a function of what else is available, which is structurally consistent with RPE computation. Adolescent VS responses, by contrast, appear to encode reward in a more absolute and context-insensitive manner, suggesting that the RPE computation itself is less sophisticated developmentally.

This finding has direct implications for risk-taking behaviour. If adolescents evaluate rewards in absolute rather than relative terms, they may overvalue uncertain options with large potential payoffs without properly weighing them against available alternatives. This is the pattern that Paulsen et al. (2012) associate with elevated risk-seeking across development, and it provides a plausible neural mechanism for why adolescents disproportionately choose high-variance options even when expected values are equivalent.

The Neural Behavioural Link: VS Differentiation and Impulsivity

The correlation findings from Table 1 of Vaidya et al. (2013) provide the most direct evidence connecting VS RPE signalling to behavioural outcomes. After controlling for age group, lower VS differentiation between the \$1.00 and \$0.20 reward conditions in the right VS was significantly associated with higher attentional impulsivity (A_Imp: $r = -0.38$, $p < 0.05$) and higher motor impulsivity (S_Imp: $r = -0.34$, $p < 0.05$). Since these correlations remained true after controlling for age group is important because it implies that the relationship between neural differentiation and impulsivity is not a proxy for being young. Instead, regardless of whether a participant is an adolescent or adult, those who show less VS sensitivity to reward magnitude differences also tend to score higher on impulsivity measures. This is consistent with the dual-systems model's prediction that the VS RPE system plays a direct role in impulse control, not simply through correlation.

The finding that adolescents scored significantly higher on attentional impulsivity than adults ($t(34) = -2.50, p = 0.02$) summarises that adolescents show weakened VS differentiation, that weakened VS differentiation predicts higher impulsivity across both groups, and that adolescents independently score higher on impulsivity. Although this does not establish causation since the cross-sectional design prevents causal inference, the finding is consistent with the account in which developmental immaturity of VS RPE signalling contributes to elevated impulsivity and, by extension, to risk-taking behaviour.

Consistency with the Dual-Systems Model

Taken together, these three findings are broadly consistent with Steinberg's (2008) dual-systems model. The model predicts that adolescent behaviour is driven by a reward system that is hypersensitive and poorly regulated by a prefrontal cortex system that has not yet fully matured. The present data partially support the first half of this prediction, though with an important nuance. The adolescent VS is not simply hyperactive; instead, it is less differentiated and less context-sensitive, suggesting a qualitatively different mode of operation rather than a quantitatively amplified one. This aligns more closely with Casey et al.'s (2008) characterisation of non-linear developmental trajectories in reward processing than with a simple hyperactivity account. The whole-brain findings support the prefrontal component of the dual-systems model. Adults showed significant activation in the anterior cingulate cortex and insula during absolute reward anticipation, regions associated with cognitive control, while adolescents showed no regions surviving the corrected threshold. This suggests that the adult reward response is accompanied by greater regulatory engagement, which is the pattern predicted by the dual-systems model.

Addressing the Bjork Inconsistency

It is important to acknowledge that not all findings in the literature are consistent with the pattern observed here. Bjork et al. (2010) reported reduced rather than elevated VS activation during reward anticipation in adolescents relative to adults on a standard MID task, a finding that appears to contradict both the dual-systems model and the present results. This inconsistency is best understood in terms of task design differences. Bjork et al. used a standard MID paradigm measuring anticipatory activation to a single reward magnitude, while Vaidya et al. (2013) examined sensitivity across multiple magnitudes and included relative reward conditions. The present findings suggest that the relevant developmental difference is not in the absolute level of VS activation but in its sensitivity to reward magnitude differences and contextual recalibration. Under a single-magnitude design, this difference would not be detectable, which may explain why Bjork et al. found reduced rather than differentiated activation. This interpretation suggests that the question of whether adolescents show more or less VS activation than adults may be less meaningful than the question of whether they show more or less differentiated VS activation, a distinction that has implications for how future studies should be designed.

Real World Implications

The pattern of these findings carries meaningful implications beyond research purposes. Spear (2000) argued that heightened reward sensitivity during adolescence lowers the threshold for drug reinforcement, and the present neural evidence provides a mechanism for this: an RPE system that does not adequately differentiate between reward magnitudes or recalibrate based on context would be poorly equipped to learn from the consequences that typically discourage escaping substance use. Similarly, Reyna and Farley's (2006) observation that adolescents rely more heavily on outcome-based reasoning than on gist-based heuristics is consistent with the absolute reward-encoding pattern observed here. Adolescents appear to process each reward

outcome individually rather than within a broader value framework, which may explain why they are more susceptible to vivid, immediate payoffs at the expense of longer-term risks. Ernst et al.'s (2006) triadic model further contextualises these findings by situating altered VS RPE-related processing within a broader motivational system in which approach drives are elevated, and regulatory capacity is reduced. This may contribute to the behavioural profile of adolescent risk-taking observed across real-world contexts, from substance usage to unsafe driving.

LIMITATIONS:

One of the major limitations of the present study is the generalisability of the sample data. Vaidya et al. (2013) recruited exclusively from the University of Iowa community and did not report racial or ethnic composition, raising concerns about WEIRD bias. Additionally, the relatively small Vaidya sample size ($n = 36$) may reduce statistical power and limit the stability of developmental comparisons. By contrast, the Smith et al. (2024) sample included 40.4% non-White participants across a broader geographic and demographic range, suggesting that lifespan demographic patterns may generalise beyond predominantly White Western samples. However, since neural findings in this study derive solely from Vaidya, the generalisability of the RPE signalling results specifically remains constrained. A primary limitation of the findings concerns data accessibility. Raw participant-level fMRI data were not publicly available for Vaidya et al. (2013), and the Smith et al. (2024) dataset required specialist neuroimaging preprocessing software beyond the scope of this study. Consequently, neural analyses were based on extracted published statistics rather than original computation, preventing the calculation of novel effect sizes or inferential statistics. Furthermore, since raw data cannot be used, it is impossible to calculate effect sizes for a novel analysis. Also, the three-level age group variable

(adolescent, young adult, older adult) combines data from 2 different studies, paradigms and methods, which means the between-group comparison is confounded by the sources. In addition to the analytical limitations, there are also several theoretical limitations of these findings. First, the dual-systems model is the dominant framework in this study, but it is not without criticism. Some researchers argue that it oversimplifies adolescent brain development and ignores individual differences. Moreover, this study focuses on group-level neural differences, but individual variation in VS activation is substantial because two adolescents can have entirely different reward sensitivity profiles. Since the research conducted is unable to distinguish whether the VS differences observed cause risk-taking or simply correlate with it, it is unclear whether this implies a direction of causation. Another very important thing to note is that findings from a controlled laboratory MID task may not generalise to the real-world risk-taking environments, where social context, emotion, and peer influence play major roles in human behaviour. Next, there are also scope limitations to these findings. Since only the ventral striatum is analysed, other regions critical to RPE signalling, such as the orbitofrontal cortex, anterior cingulate, and ventral tegmental area, are not examined. Furthermore, only monetary reward is considered, although adolescent risk-taking in other peer contexts may involve different neural mechanisms entirely. Finally, this study covers only adolescents vs. young adults; middle-aged and early adolescence (ages 8-12) are absent, which matters for understanding the onset of the developmental trajectory.

CONCLUSION:

Connecting back to the research question: to what extent do age-related differences in ventral striatal RPE signals explain elevated risk-taking behaviour in adolescents compared to adults, the present findings support a partial and affirmative answer. Developmental differences

in VS reward magnitude sensitivity, relative reward recalibration, and the independent association between VS differentiation and impulsivity are consistent with the hypothesis that adolescent risk-taking is heavily associated with an immature RPE signalling system. However, the causal direction of this relationship cannot be established from the available cross-sectional data because the findings derive from a single primary source with a small sample, and the translation from laboratory reward sensitivity to real-world risk-taking remains inferential. Future research employing longitudinal designs, larger and more diverse samples, and paradigms that more closely approximate real-world decision contexts would be needed to establish the strength and generalisability of this mechanism more accurately. Understanding how developmental reward-learning systems shape behaviour may ultimately contribute to more targeted interventions for adolescent decision-making, behavioural regulation, and mental health outcomes. These neural findings carry substantial implications for adolescent financial decision-making, in which RPE-driven behaviour has well-documented real-world consequences. The weakened VS magnitude differentiation observed in adolescents suggests that proportionally larger financial stakes may not generate proportionally stronger neural reward signals. Within a reinforcement learning framework, this is significant because the feedback loop through which individuals learn to treat high-stakes decisions with greater caution depends on the RPE system encoding magnitude differences accurately. If the adolescent VS does not scale its response proportionally across reward sizes, the experimental learning that ordinarily produces risk calibration over time may be impaired. This is consistent with Cauffman et al.'s (2010) finding that adolescents show poorer decision-making than adults on the Iowa Gambling Task, a paradigm in which adaptive performance requires learning to avoid high-variance, high

penalty options. This is precisely the type of magnitude-sensitive updating that the present findings suggest is developmentally immature in adolescents.

The context-insensitive reward encoding pattern extends this further. Adults in Vaidya et al. (2013) recalibrated their VS response to the same objective reward depending on what alternatives were available, a pattern consistent with the comparative value computations that underlie rational financial reasoning. However, adolescents showed no such recalibration, encoding each reward outcome in absolute rather than relative terms. In financial contexts, this distinction is highly consequential because the same nominal gain or loss carries a different meaning depending on total stake size, available alternatives, and opportunity cost. All of these require the kind of contextual reward comparison that the adolescent VS appears not yet to perform. Reyna and Farley's (2006) observation that adolescents rely more heavily on outcome-based reasoning than on gist-based heuristics is consistent with this absolute encoding pattern, suggesting that vivid, immediate payoffs may carry disproportionate neural prominence relative to longer-term or probabilistic consequences. Together, reduced magnitude sensitivity and context-insensitive encoding provide a potential neural mechanism through which developmental RPE immaturity may contribute to the elevated financial risk-taking observed in adolescents, though longitudinal research linking VS differentiation directly to real-world financial behaviour would be needed to establish this relationship more strongly.

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